Farjin Anyar Rahman

farjin.358@gmail.com | +880 1921 260277 | Sylhet, Bangladesh

EDUCATION

B.Sc. (Engg.) in Industrial and Production Engineering 2022 – 2026
Shahjalal University of Science and Technology (SUST), Sylhet, Bangladesh (Session 2020–21)
CGPA: 3.91 / 4.00; **First Class First with Distinction**
Thesis: *Physics-Based Non-Invasive Abdomen Fat Estimation Using Skin Surface Temperature: A Finite Element Forward and Inverse.* Supervisor: Dr. Abul Mukid Md. Mukaddes (IPE, SUST)

Higher Secondary Certificate (Science) 2019 – 2021
SOS Hermann Gmeiner College, Dhaka | GPA: 5.00 / 5.00

Secondary School Certificate (Science) 2017 – 2019
SOS Hermann Gmeiner College, Dhaka | GPA: 5.00 / 5.00

PUBLICATIONS

Conference Papers

- **Thermographic Signature of a Subcutaneous Lipoma: A Finite Element Bio-Heat Transfer Analysis of the Abdominal Skin Temperature**. *10th BSME International Conference on Thermo-Fluid Engineering (ICTE 2026), BUET, Dhaka, December 2026*.
  Authors: Farjin Anyar Rahman, Khandakar Ikramul Islam, Saifur Rahman Tashki, Ashraful Huq and Prof. Dr. Abul Mukid Md. Mukaddes.

- **Thermographic Signature of Abdominal Adiposity: A Finite Element Bio-Heat Transfer Analysis of the Skin Surface Temperature**. *International Conference on Mechanical, Industrial and Energy Engineering (ICMIEE 2026), KUET, Khulna, December 2026.*
  Authors: Farjin Anyar Rahman, Saifur Rahman Tashki, and Prof. Dr. Abul Mukid Md. Mukaddes.

- **SHARP-GNN: A Time-Aware Relational Graph Neural Network for Joint Prediction of Shortage Magnitude and Onset Risk in Pharmaceutical Supply Chain**. *Conference on Building Resilient Supply Chains (SCM 2026), East West University, Dhaka, December 2026*.
  Authors: Farjin Anyar Rahman, Md. Mahfuz Ahmed, Prof. Dr. Choudhury Abul Anam Rashed, and Prof. Dr. Md. Ariful Islam.

- **Modelling the Interdependence of Sustainable Supply Chain Risk Factors in the Petroleum Industry Under Crisis Conditions: An ISM–TISM Approach**. *Conference on Building Resilient Supply Chains (SCM 2026), East West University, Dhaka, December 2026*.
  Authors: Nahid Hasan Nayem, Farjin Anyar Rahman, Mahamudul Hashan, Md. Minhazul Abedin, Labiba Nusrat Jahan, and Prof. Dr. Choudhury Abul Anam Rashed

- **Relational GNN for Joint Prediction of Shortage Magnitude and Onset Risk in Pharmaceutical Supply Chain**. *2nd International Student Research Conference (ISRC) 2026, Dhaka University, Dhaka, December 2026*. [Abstract]
  Authors: Md. Mahfuz Ahmed, Farjin Anyar Rahman, and Prof. Dr. Choudhury Abul Anam Rashed.

- **Non-Invasive Abdominal Fat Estimation from Skin Surface Temperature: A Finite Element Approach**. *6th Young Scientist Congress, Bangladesh Academy of Sciences, Dhaka, April 2026*. [Abstract] *Best Paper Oral Presentation*, 2nd Place (Mathematical, Engineering and Computer Sciences).
  Authors: Farjin Anyar Rahman, Saifur Rahman Tashki, and Prof. Dr. Abul Mukid Md. Mukaddes.

- **Development of a Computational Model of Human Abdomen to Analyze Bioheat Transfer and Its Correlation with Fat Distribution**. *International Conference on Science, Technology, Engineering, Mathematics, and Education (iSTEME2025 2025), Huston-Tillotson University, Austin, Texas, December 2025*. [Abstract]
  Authors: Farjin Anyar Rahman, Saifur Rahman Tashki, and Prof. Dr. Abul Mukid Md. Mukaddes.

- **Scope of Ergonomics: Case Study of Academia and Industry in Bangladesh**. *10th North American International Conference on Industrial Engineering and Operations Management (IEOM), Orlando, Florida, June 2025*.
  Authors: Prof. Dr. Mohammad Iqbal, Prof. Dr. Salma Akhter, Farjin Anyar Rahman, Sadia Anjum, and Tasnim Iqbal.

- **Ergonomics in Industrial Design**. *8th International Conference on Mechanical, Industrial and Energy Engineering (ICMIEE 2024), KUET, Khulna, January 2025*.
  Authors: Prof. Dr. Mohammad Iqbal, Prof. Syed Misbah Uddin, Farjin Anyar Rahman, Syed Sabbir Hasan, Md. Mushtaq Tahmid, and Achia Khatun

Journal Manuscripts

- **Numerical Bioheat Modelling Connection With Biology: A Systematic Review of a Shrinking Intersection** *[Work In Progress]*
  Authors: Prof. Dr. Abul Mukid Md. Mukaddes, Farjin Anyar Rahman, and Saifur Rahman Tashki.

- **Physics-Based Non-Invasive Abdomen Fat Estimation Using Skin Surface Temperature: A Finite Element Forward and Inverse Study** *[Work In Progress]*
  Authors: Prof. Dr. Abul Mukid Md. Mukaddes, Farjin Anyar Rahman, and Saifur Rahman Tashki.

- **From Tabular to Relational: A Systematic Review of Machine & Graph Learning in Pharmaceutical Supply Chains** *[Work In Progress]*
  Authors: Prof. Dr. Choudhury Abul Anam Rashed, Farjin Anyar Rahman, and Md. Mahfuz Ahmed.

- **Testing the Shape of the Digitalization-Human Comfort Relationship: Meta-Analytical Evidence for an Asymmetric Plateau-and-Cliff** *[Work In Progress]*
  Authors: Prof. Dr. Mohammad Iqbal, Farjin Anyar Rahman, Mst. Fatiha Akhter Chowdhury, and Md. Mahfuz Ahmed.

RESEARCH EXPERIENCE

Research Student, Bioheat Transfer and Inverse Modelling 2025 – Present
Department of IPE, SUST (supervised by Dr. A. M. M. Mukaddes)
- Modelled steady-state Pennes bioheat transfer in a abdominal cross-section in COMSOL Multiphysics for slim and obese subjects. Validated it against experimental thermography.
- Solved the inverse problem of recovering abdominal fat area from skin-surface temperature.
- Simulated a subcutaneous lipoma in a multi-tissue torso cross-section and reproduced the physiologically expected "thermal dip" surface signature.

Research Student, Machine Learning for Pharmaceutical Supply Chains 2026 – Present
Independent research project, SUST (supervised by Dr. C. A. A. Rashed)
- Developed SHARP-GNN, a time-aware relational graph attention network (GRU temporal encoder, six relation types, dual output heads, about 34.6k parameters).

Research Student, Ergonomics and Human-Computer Interaction 2024 – Present
Independent research project, SUST (supervised by Dr. M. Iqbal)
- Conducted study on ergonomics in industrial design and on the scope of ergonomics in Bangladeshi academia and industry.
- Meta-analysis testing the shape of the digitalization–human comfort relationship and human artificial intelligence interaction [HAII].

REVIEWING AND TECHNICAL SERVICE

Reviewer and Technical Committee Member June 2026
11th North American IEOM Conference, Milwaukee, Wisconsin

SCHOLARSHIPS AND HONORS

- SUST-STEM Professors Roy-Mondal-Miah Memorial Scholarship. October 2024
- Award of Excellence, IPE Association, SUST. October 2023
- Future Scholars Celebration '23, Student Aid SUST. August 2023
- Department Merit Scholarship, SUST. 2025 – 2026

AWARDS AND RECOGNITIONS

- Best Paper Oral Presentation, 2nd Place (Mathematical, Engineering and CS), BAS. April 2026
- Finalist, IPESphere Project Fair (EcoGrind), SUST IPE Alumni Association. February 2025
- Champion, Mastermind 2.0 National Case Competition, SUST Career Club. December 2024
- Champion, Inter-Department Presentation Competition (IDPC 2.0), SUST Speakers' Club. December 2024
- Top Three, Case Quest 1.0, IPE Association, SUST. 2024

CERTIFICATIONS

BDRS Research Learning and Paper Writing Program, Season-2, Bangladesh Research Society Dec 2025
Credential ID: BDRS.RLPW.02.2024-25.12.30.096

Basic Data Analysis Course ON STATA 2024, Bangladesh Research Society Feb 2025

INDUSTRY EXPERIENCE

Industrial Trainee, JT International Bangladesh Limited May 2026
One-month industrial attachment, Make-Pack (Cigarette and Filter Production); TPM and SAP documentation.

TEACHING EXPERIENCE

IELTS and IGCSE/IAL Tutor (Mathematics, Physics, English), Private 2020 – Present

IELTS Examiner, Mentors', Sylhet Apr 2022 – Dec 2022

IELTS Instructor, TENs English, Sylhet 2022 – 2023

LEADERSHIP AND SERVICE

Ergonomics Society of SUST (ESS), SUST **Sep 2025 – Present**
- *Presidium Coordinator* Jun 2026 – Present
- *Founding President* Sep 2025 – Jun 2026

Sylhet Debate Federation, Sylhet **Feb 2024 – Present**
- *Presidium Member* Oct 2026 – Present
- *Assistant General Secretary* Feb 2024 – Sep 2026

SUST School of Debate (SUST-SD), SUST **Sep 2022 – Nov 2025**
- *President* Mar 2025 – Nov 2025
    - Convener, "SUST Open 2025" Nov 2025
    - Session Chair, "Ramadan Blitz 2025" Mar 2025
- *Debate Director* Mar 2024 – Mar 2025
    - Convener, "Ramadan Blitz 2024" Mar 2024
- *Assistant Debate Director* Mar 2023 – Mar 2024
- *Debater* Sep 2022 – Nov 2025

National Student Data Corps SUST, SUST **Feb 2025 – Mar 2026**
- *Assistant General Secretary* Feb 2025 – Mar 2026

TEDx SUST, SUST **Jul 2025 – Sep 2025**
- *Head of Curation* Jul 2025 – Sep 2025

IPE Association, SUST, SUST **Mar 2022 – Jan 2024**
- *Publication Secretary* Feb 2023 – Jan 2024
- *Executive Committee Member* Mar 2022 – Feb 2023

SUST Career Club, SUST **Oct 2022 – Jan 2023**
- *Departmental Executive* Oct 2022 – Jan 2023

DEBATE EXPERIENCE

Debating Achievements:

- **Runners-Up, 2nd Breaking Team and Debater of the Tournament**, 2nd LUDC Divisional Debate Fest 2024, Leading University Debating Club (LUDC) Sep 2024
- **Novice Finalist and 3rd Breaking Team**, AUW Marchentity 2024,AUWDS. Mar 2024
- **Quarter-Finalist and 6th Breaking Team**, JUDO National Debate Festival 2023, JUDO Oct 2023
- **Semifinalist and 4th in Tabs**, Communication in Action (CIA) Season 5, CUFBDA Oct 2023

Adjudication Experience:

- **Core Adjudicator,** Parliamentary Fiesta 1.0, Blue Bird High School, and College Debating Club. Apr 2026
- **Grand Finale Panelist**, Inter-Departmental Parliamentary Debate Competition 2025, MUDC. Nov 2025
- **Top Breaking Adjudicator**, SUST Open 2025, SUST School of Debate. Nov 2025
- **Grand Finale Chair**, First MUDC Intra Debating Championship, MUDC. Oct 2025
- **Core Adjudicator**, SDF Divisionals 2025, Sylhet Debate Federation. Oct 2025
- **Grand Finale Panelist (Core Adjudicator)**, SUST Divisionals 2025, SUST School of Debate. Aug 2025
- **Independent Adjudicator**, Oxford WSDC 2025, Oxford Debating Society. Jul 2025
- **Independent Adjudicator**, Austria Pre-WSDC 2025. Jun 2025
- **Grand Finale Panelist (Adjudicator)**, The Great Phoenix War, SUST School of Debate. Feb 2025
- **Independent Adjudicator**, Capitalism IV 2024, Melbourne University Debating Society. Nov 2024
- **Independent Adjudicator**, Monash Pre-Australs 2024, Monash Association of Debaters. Jun 2024
- **Independent Adjudicator**, Penn WSDC 2024, Penn Debate Society. 2024
- **Independent Adjudicator**, Princeton Online WSDC 2024. 2024
- **Independent Adjudicator,** Beacon of Hope: Save Farhan Fundraiser, BUP Debate Club. Apr 2024
- **Top Breaking Trainee Judge**, Alpha Novice Debate Cup, Argue Well. Feb 2024
- **Breaking Adjudicator**, Chattogram BP, Chattogram Cantonment Public College Debating Society. Jan 2024
- **Trainee Adjudicator**, JUDO National Debate Festival 2023 (Inter-College), JUDO. Oct 2023
- **Adjudicator**, Inter-Departmental Debate Competition, Department of Bangla, SUST Oct 2023

Debate Trainer and Mentor (Voluntary): SUST-SD, SAUDS, MUDC, BBHSCDC, SGCDC, SDF.

PROJECTS

Pressure Control Valve Jun 2025 – Sep 2025
(with Dewan Reza Hamid, Anisur Rahman, Sabid Al Haque, Sawda Begum and Anindita Ghosh) Designed and fabricated a floating-ball pressure control valve, followed by force and pressure analysis of its performance.

Digital Grain Moisture Meter Nov 2024 – Mar 2025
(with Mst. Fatiha Akter Chowdhury, Nur Muhammad Ashik and Prity Rani Sarker) Developed a portable, low-cost on-site moisture meter to help small-scale farmers and food producers in Bangladesh reduce post-harvest losses.

Quick Composter May 2024
(with Dewan Reza Hamid, Anisur Rahman and Arnob Roy) Designed a compact firing-and-grinding composter converting organic waste to compost in 4–5 hours.

EcoGrind, IPESphere Project Fair, SUST IPE Alumni Association Feb 2025
(with Dewan Reza Hamid Karzai) Finalist.

SKILLS

Simulation: COMSOL Multiphysics (Heat Transfer, Bioheat), MATLAB. [FEM; Inverse/Computational Estimation]

Programming: Python, C, C++.

Statistics And Design: Arena, AutoCAD, MIRO, SPSS, Design Expert, STATA, Sigmaplot.

Research And Leadership: Research Skills, Data Analysis, Publication Writing, Event Management, Educational Leadership, Convening, Adjudication.

Languages: English (Fluent, IELTS 8.0); Bangla (Native).